

Supplement to “Silent Aliasing in Fixed Fourier-Feature Coordinate Models”: Full Proofs

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Numbering follows the main paper (T1–T4 are Theorems 1–4 there). Setting and notation as in the main paper: $e_\omega(x) = e^{i2\pi\omega x}$ on $[0, 1)$, model set $\Lambda = \{\omega_1, \dots, \omega_m\}$, samples $T = \{t_j\}_{j=1}^N$, synthesis matrix $\Phi_{jk} = e_{\omega_k}(t_j)$, sample atom $\phi_\nu = (e^{i2\pi\nu t_j})_j$, visibility $v_T(\nu) = \|(\mathbf{I} - \mathbf{P}_\Lambda)\phi_\nu\|/\sqrt{N}$, aliasability $a_T(\nu) = \|\Phi^\dagger \phi_\nu\|$. Every constant below is also asserted numerically in `tests/test_identifiability.py`.

1 Proof of Theorem 1 (exact equivalence on grid subsets)

(i) $v_T(\nu) = 0$ iff $(\mathbf{I} - \mathbf{P}_\Lambda)\phi_\nu = 0$ iff $\phi_\nu \in \text{range } \Phi$, which is precisely the statement that some coefficient vector $\mathbf{c}$ has $\Phi\mathbf{c} = \phi_\nu$, i.e. the unit tone at ν agrees with a model element at every sample. For the grid statement, let $t_j = g_j/Q$ with $g_j \in \{0, \dots, Q-1\}$ and $\nu = \omega_k + rQ$, $r \in \mathbb{Z}$. Then

$$e^{i2\pi\nu g_j/Q} = e^{i2\pi\omega_k g_j/Q} e^{i2\pi r g_j} = e^{i2\pi\omega_k g_j/Q},$$

since $rg_j \in \mathbb{Z}$; hence $\phi_\nu = \phi_{\omega_k}$ entrywise, for *every* subset of the grid. Consequently $v_T(\nu) = 0$; moreover, because $\sigma_{\min}(\Phi) > 0$ makes the LS solution unique, the fit of $y = \phi_\nu$ is the unique $\mathbf{c}$ with $\Phi\mathbf{c} = \phi_{\omega_k}$, namely the k -th unit vector, and the residual is exactly zero.

(ii) By (i), $S_T f_1 = S_T f_2$ pointwise. For any noise mechanism that acts on $S_T f$ (additive or otherwise, as long as its law depends on the signal only through $S_T f$), the observation distributions under f_1 and f_2 are therefore identical. Let $\hat{f} = \delta(\mathbf{y})$ be any estimator. For every realization $\mathbf{y}$,

$$\|\delta(\mathbf{y}) - f_1\|_{L^2} + \|\delta(\mathbf{y}) - f_2\|_{L^2} \geq \|f_1 - f_2\|_{L^2} = |a| \|e_\nu - e_{\omega_k}\|_{L^2} = |a|\sqrt{2},$$

using the triangle inequality and, for the last equality, that e_ν and e_{ω_k} are distinct integer frequencies, hence orthonormal in $L^2[0, 1)$. Thus $\max_i \|\delta(\mathbf{y}) - f_i\| \geq |a|/\sqrt{2}$ pointwise (a two-point bound in the sense of Le Cam, here even realization-wise).

(iii) If $[\Phi \phi_\nu]$ has full column rank then $\phi_\nu \notin \text{range } \Phi$, so $v_T(\nu) > 0$ by (i). $\square$

Remark 1. *The full-column-rank hypothesis on Φ requires $N \geq m$ and the elements of Λ to be distinct modulo Q (two atoms congruent mod Q have identical columns on any grid subset). “Any N ” claims are always to be read as “any grid-subset size N for which this rank condition holds”.*

2 Proof of Theorem 2 (Riesz conversion and decomposition)

(i) G_{L^2} is the Gram matrix of $\{e_{\omega_k}\}$ in $L^2[0, 1)$: $\|\sum_k c_k e_{\omega_k}\|_{L^2}^2 = \mathbf{c}^* G_{L^2} \mathbf{c}$, and the Rayleigh quotient of the Hermitian PSD matrix G_{L^2} lies in $[\lambda_{\min}, \lambda_{\max}]$. The entries are $\int_0^1 e^{i2\pi(\omega_\ell - \omega_k)t} dt = \frac{e^{i2\pi\delta} - 1}{i2\pi\delta} = e^{i\pi\delta} \frac{\sin \pi\delta}{\pi\delta}$ for $\delta = \omega_\ell - \omega_k \neq 0$, and 1 on the diagonal. G_{L^2} is singular iff the atoms are linearly dependent in L^2 , which for finitely many distinct frequencies never happens (distinct exponentials are linearly independent); hence $\lambda_{\min} > 0$ for distinct frequencies.

(ii) Expand $\hat{f} - f = \sum_k \Delta c_k e_{\omega_k} - \sum_l a_l e_{\nu_l}$ and evaluate $\|\hat{f} - f\|_{L^2}^2$ as the quadratic form of the joint Gram matrix of $(\{\omega_k\}, \{\nu_l\})$ applied to $(\Delta\mathbf{c}, -\mathbf{a})$; grouping blocks gives Eq. (3) of the main paper. For all-integer frequencies the off-diagonal blocks vanish (orthonormality), killing the cross term. $\square$

3 Proof of Theorem 3 (random sampling breaks exact aliasing)

We use the following complex Hoeffding bound throughout: if $Z_1, \dots, Z_N$ are independent with $|Z_j| \leq 1$, then for $\bar{Z} = \frac{1}{N} \sum_j Z_j$,

$$\Pr(|\bar{Z} - \mathbb{E}\bar{Z}| \geq \varepsilon) \leq 4 \exp(-N\varepsilon^2/4), \quad (1)$$

obtained by applying the real Hoeffding inequality to the real and imaginary parts (each in $[-1, 1]$) with radius $\varepsilon/\sqrt{2}$ and a union bound.

3.1 Part (a): jitter

Let $t_j = g_j/Q + \eta_j$ with g_j grid points and η_j i.i.d. mean-zero with characteristic function χ_η , and $\nu - \omega_k = rQ$, $r \in \mathbb{Z} \setminus \{0\}$. The fold coherence is

$$C = \frac{1}{N} \sum_j e^{i2\pi(\nu - \omega_k)t_j} = \frac{1}{N} \sum_j \underbrace{e^{i2\pi r g_j}}_{=1} e^{i2\pi r Q \eta_j} = \frac{1}{N} \sum_j e^{i2\pi r Q \eta_j},$$

an empirical characteristic function with $\mathbb{E}C = \chi_\eta(2\pi r Q)$ and concentration (1): $|C - \chi_\eta(2\pi r Q)| \leq 2\sqrt{\log(4/\delta)/N}$ with probability $\geq 1 - \delta$. For Gaussian jitter, $\chi_\eta(u) = e^{-u^2 \sigma_t^2/2}$, so $\chi_\eta(2\pi r Q) = e^{-2(\pi r Q \sigma_t)^2}$; for uniform jitter of width w , $\chi_\eta(2\pi r Q) = \text{sinc}(rQw)$.

From coherence to visibility. Let $\mathbf{b} = \Phi^* \phi_\nu / N \in \mathbb{C}^m$ and $\mathbf{G}_N = \Phi^* \Phi / N$. Then

$$v_T(\nu)^2 = \frac{1}{N} \|\phi_\nu\|^2 - \frac{1}{N} \phi_\nu^* \mathbf{P}_\Lambda \phi_\nu = 1 - \mathbf{b}^* \mathbf{G}_N^{-1} \mathbf{b}.$$

Assume additionally—as in the theorem statement—that the grid points g_j are drawn i.i.d. uniformly from the grid (a uniformly random subset drawn without replacement obeys the same bounds with the same constants by Serfling’s inequality for sampling without replacement) and that the elements of Λ are distinct modulo Q (otherwise two Gram columns coincide on every grid subset and $\mathbf{G}_\infty \neq \mathbf{I}$, breaking the perturbation step below). Each entry of $\mathbf{b}$ other than the fold entry, and each off-diagonal entry of $\mathbf{G}_N$, is an empirical mean of independent unit-modulus terms whose expectation vanishes (distinct residues modulo Q average to zero over the uniform grid, and the jitter factor only multiplies by a number of modulus ≤ 1); the fold entry of $\mathbf{b}$ is C above. By (1) and a union bound over these $m + m^2$ quantities, with probability $\geq 1 - \delta$ all of them are within $\varepsilon = 2\sqrt{\log(4(m + m^2)/\delta)/N}$ of their means. Writing $\mathbf{G}_N = \mathbf{I} + \mathbf{E}$ with $\|\mathbf{E}\| \leq m\varepsilon$ and $\mathbf{b} = C\mathbf{e}_k + \mathbf{u}$ with $\|\mathbf{u}\| \leq \sqrt{m}\varepsilon$, a first-order expansion of $\mathbf{b}^* \mathbf{G}_N^{-1} \mathbf{b}$ gives

$$|v_T(\nu)^2 - (1 - |\chi_\eta(2\pi r Q)|^2)| \leq c m \varepsilon$$

for an absolute constant c (valid while $m\varepsilon < 1/2$), which is the $O_P(m\sqrt{\log(m/\delta)/N})$ term of the theorem. The small-jitter law follows from $1 - e^{-2u} \leq 2u$ and $1 - e^{-2u} \rightarrow 2u$ as $u \rightarrow 0$ with $u = (2\pi r Q \sigma_t)^2/2$: $v_T(\nu) = \sqrt{1 - \chi^2} + O(m\varepsilon) = 2\pi|r|Q\sigma_t(1 + o(1))$.

3.2 Part (b): i.i.d. samples

Entries of $\mathbf{G}_N - \mathbf{G}_\infty$ and of $\mathbf{b}_\nu = \Phi^* \phi_\nu / N$ (for every $\nu \in \Omega$) are empirical means of independent unit-modulus variables; by (1) and a union bound over the $m^2 + mK$ of them, with probability $\geq 1 - \delta$ every such quantity is within $\varepsilon = 2\sqrt{\log(4(mK + m^2)/\delta)/N}$ of its limit. Under the hypotheses, the limiting coherences vanish, so $\|\mathbf{b}_\nu\| \leq \sqrt{m}\varepsilon$ for all $\nu \in \Omega$, and $\|\mathbf{G}_N - \mathbf{G}_\infty\| \leq \|\cdot\|_F \leq m\varepsilon$, hence $\lambda_{\min}(\mathbf{G}_N) \geq \lambda_{\min} - m\varepsilon > 0$. Therefore, uniformly over Ω ,

$$a_T(\nu) = \|\mathbf{G}_N^{-1} \mathbf{b}_\nu\| \leq \frac{\|\mathbf{b}_\nu\|}{\lambda_{\min}(\mathbf{G}_N)} \leq \frac{\sqrt{m}\varepsilon}{\lambda_{\min} - m\varepsilon}. \quad \square$$

Remark 2. The bound is a union bound and is loose by design (Fig. 1b of the main paper); its role is the rate $N^{-1/2}$ and the uniformity over the candidate set, both matched empirically. It is finite exactly when $m\varepsilon < \lambda_{\min}$ and is reported as vacuous otherwise, never silently extrapolated.

4 Proof of Theorem 4 (detection dichotomy)

(a) If $S_T f_1 = S_T f_2$ and the observation law depends on the signal only through $S_T f$ (as for additive noise $\mathbf{y} = S_T f + \varepsilon$ with ε independent of f), the distributions of $\mathbf{y}$ under the two hypotheses are equal. For any (possibly randomized) test $\varphi(\mathbf{y}) \in [0, 1]$, $\mathbb{E}_{f_1} \varphi = \mathbb{E}_{f_2} \varphi$: power equals size.

(b) Under $\mathbf{y} = \mathbf{D}\boldsymbol{\beta} + \mathbf{s} + \varepsilon$ with $\varepsilon \sim \mathcal{N}(0, \sigma^2 \mathbf{I})$ and $\mathbf{P}_D$ the projector onto the rank- p column space of $\mathbf{D}$, $(\mathbf{I} - \mathbf{P}_D)\mathbf{y} = (\mathbf{I} - \mathbf{P}_D)(\mathbf{s} + \varepsilon)$ is Gaussian with mean $(\mathbf{I} - \mathbf{P}_D)\mathbf{s}$ and covariance $\sigma^2(\mathbf{I} - \mathbf{P}_D)$; hence $\|(\mathbf{I} - \mathbf{P}_D)\mathbf{y}\|^2/\sigma^2$ is noncentral χ^2 with $N - p$ degrees of freedom and noncentrality $\lambda = \|(\mathbf{I} - \mathbf{P}_D)\mathbf{s}\|^2/\sigma^2$ (central when $\mathbf{s} = 0$ or when the component is coherent, $(\mathbf{I} - \mathbf{P}_D)\mathbf{s} = 0$). The stated power expression is the definition of the noncentral- χ^2 survival function at the central $(1-\alpha)$ -quantile. $\square$

5 Background: average-case decomposition

Proposition S 1 (average-case LS error; standard bookkeeping). *Let $\mathbf{y} = \boldsymbol{\Phi}\mathbf{c}^* + \boldsymbol{\Phi}_{\text{out}}\mathbf{a} + \varepsilon$ with $\sigma_{\min}(\boldsymbol{\Phi}) > 0$, the a_l zero-mean, uncorrelated, $\mathbb{E}|a_l|^2 = p_l$, independent of ε , and $\mathbb{E}[\varepsilon\varepsilon^*] = \sigma^2\mathbf{I}$. Then, with $\mathbf{M} = (\boldsymbol{\Phi}^*\boldsymbol{\Phi})^{-1}\boldsymbol{\Phi}^*$,*

$$\mathbb{E}\|\hat{\mathbf{c}} - \mathbf{c}^*\|^2 = \sigma^2 \text{tr}((\boldsymbol{\Phi}^*\boldsymbol{\Phi})^{-1}) + \text{tr}(\mathbf{M}\boldsymbol{\Phi}_{\text{out}}\text{diag}(\mathbf{p})\boldsymbol{\Phi}_{\text{out}}^*\mathbf{M}^*).$$

Proof. $\hat{\mathbf{c}} - \mathbf{c}^* = \mathbf{M}(\boldsymbol{\Phi}_{\text{out}}\mathbf{a} + \varepsilon)$; expand the squared norm; the cross term vanishes by independence and zero means; the two quadratic terms are the stated traces. $\square$

This proposition is retained for completeness only; it is standard omitted-variable bias/variance bookkeeping and is not claimed as a contribution.